

D-18/2113

DISCRETE MATHEMATICAL STRUCTURE-2305T

(Semester-III)

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt *three* question each from Section A and B.
Section C is compulsory.

SECTION-A

- I. What is meant by composition of functions? Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ? Discuss. 5
- II. What is an equivalence relation? What are the three properties that define it? Let R be the relation on the set of real numbers such that aRb if and only if $a-b$ is an integer. Is R an equivalence relation? Discuss. 5
- III. (a) Three cards are chosen one after the other from a 52-card deck. Find the number m of ways this can be done: (i) with replacement; (ii) without replacement.

(b) What does binomial coefficient represent? A farmer buys 3 cows, 2 pigs, and 4 hens from a man who has 6 cows, 5 pigs, and 8 hens. Find the number m of choices that the farmer has.

IV. What is a recurrence relation? Why is it important? Find a recurrence relation and give initial conditions for the number of bit strings of length n that do not have two consecutive 0s. Appropriately assume any required information yourself.

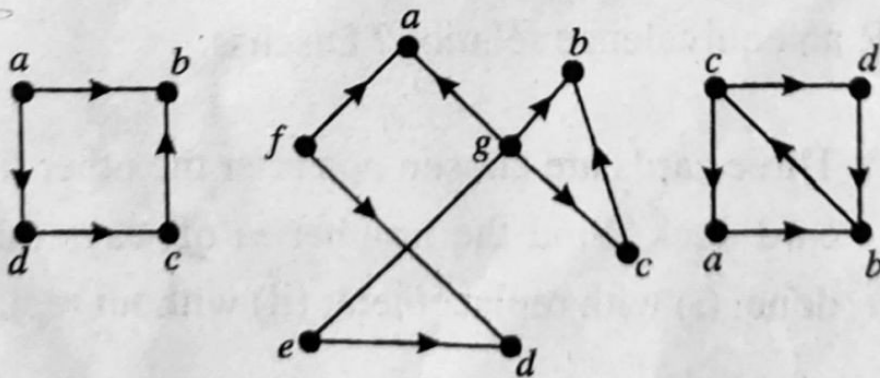
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V. What do you understand by Generating Functions? Briefly discuss their role in Combinatorics. How the concept of generating functions can be applied to find the different ways eight identical cookies can be distributed among three distinct children if each child receives at least two cookies and no more than four cookies. Appropriately assume any required information yourself.

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SECTION-B

- VI. What do you understand by Euler Circuit and Euler Path? Which of the directed graphs in the following figure have an Euler circuit? Of those that do not, which have an Euler path? Discuss. 5



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- VII. Explain Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF) in propositional logic. How do they differ? Give suitable examples. 5
- VIII. Explain any two ways to represent graphs and discuss their advantages and disadvantages. 5

- VIII. Explain any two ways to represent graphs and discuss their advantages and disadvantages. 5
- IX. Explain the concept of Monoid and Abelian Group. What are their properties? Discuss. 5

SECTION-C
(Compulsory Question)

- X. (a) Define a relation and give an example of a binary relation. 2
- (b) Provide an example of a function that does not have an inverse. 2
- (c) How does permutation differ from combination? Discuss in brief. 2
- (d) What do you understand by the Pigeonhole Principle? Discuss in brief. 2
- (e) What do you understand by Graph isomorphism? Discuss in brief. 2
- (f) What is an adjacency matrix? How is it used to represent a graph? 2
- (g) What do you understand by the terms tautology and contradiction? 2

- (h) How would you represent the statement "For all x , there exists a y such that $P(x, y)$ " using quantifiers and predicates? 2
- (i) What do you understand by Algebraic Structure? Discuss in brief. 2
- (j) What is a subgroup in the context of group theory? Discuss in brief. 2
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SECTION—A

1. Among the first 1200 positive integers :

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⑦

- (a) Determine the integers which are not divisible by 5, nor by 7, nor by 9.
- (b) Determine the integers which are divisible by 5, but not by 7, not by 9.

- ⑦ 2. Let $A = \{1, 2, 3, \dots, 9\}$ and let $\sim$ be the relation on $A \times A$ defined by : 5

$$(a, b) \sim (c, d) \text{ if } a + d = b + c.$$

Determine whether $\sim$ forms an equivalence relation or not.

- ⑦ 3. It is known that in a University 60% of professors play tennis, 50% play cricket, 70% jog, 20% play tennis and cricket, 40% play cricket and jog and

30% play tennis and jog. If someone claimed that 20% professors jog and play tennis and cricket, would you believe this claim? Justify. 5

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = ax + b$ where $a, b, x \in \mathbb{R}$ and $a \neq 0$. Determine whether f is invertible and find inverse of f (if any). 5

5. Find the complete solution of recurrence relation :

$$A(r) + 5A(r-1) + 6A(r-2) = 3r^2 - 2r + 1. \quad 5$$

SECTION—B

- ④ 6. Rewrite the following argument symbolically and determine its validity. 5

If John graduates he gets a job.

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[P. T. O.]

John does not get a job.

Therefore, John does not graduate.

- ③ 7. State and prove Euler's formula for planer graphs. 5

8. Consider $G = \{1, 5, 7, 11, 13, 17\}$ under multiplication modulo 18. 5

(a) Build the multiplication table to G .

(b) Find 5^{-1} , 7^{-1} and 17^{-1} .

(c) Find the order and group generated by :

(i) 5

(ii) 13.

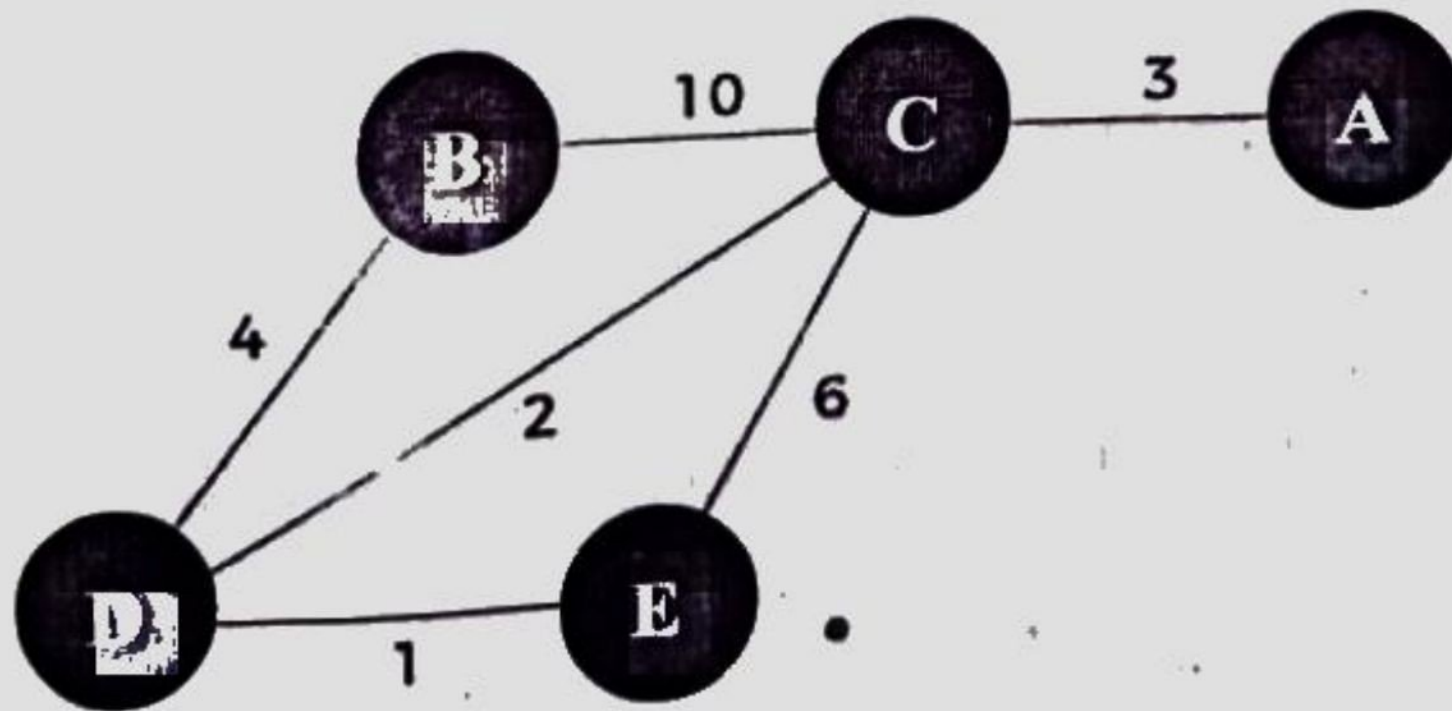
(d) Identify whether G is cyclic.

9. Identify whether the following statements are equivalent. 5

“If the farmers don’t reduce the price then the Government will take action against them.”

“It is not true that the farmers do not reduce the Prices and Government does not take action against them.”

- ③ 10. Construct MST for the following graph using Prim's algorithm. 5



SECTION—C

11. Write short notes on the following : $10 \times 2 = 20$

- ② (i) In how many ways can 5 gentlemen and 5 ladies be seated round a table so that no two ladies are together?
- ④ (ii) Determine DNF of $p \leftrightarrow (\sim p \vee \sim q)$?
- ③ (iii) Differentiate between Hamiltonian circuit and Eulerian circuit.
- ⑤ (iv) List the conditions under which two groups could be isomorphic.
- ⑥ (v) Find the sequence whose generating function is $1/(z^2 - 4)$?
- ⑦ (vi) Find the range and domain of the following :

- ①. (vi) Find the range and domain of the following :

$$f(x) = |x - 3| / x - 3.$$

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- ①. (vii) Define the POSET with suitable example.
- ②. (viii) Find the generating function if $S(n) = ba^n, n \geq 0.$
- ⑤. (ix) State Lagrange's theorem.
- ⑤. (x) Prove that identity element in a group is unique.

G-53/2114

DISCRETE MATHEMATICAL STRUCTURES

Paper-2305T

Semester-III

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. Determine whether the relation R is reflexive, symmetric, anti-symmetric or transitive, given that R is a relation on $\mathbb{Z}$, where xRy means $(x + y)$ is odd. 5

2. Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 2^n + n$, with initial conditions given that $a_0 = 1, a_1 = 1$. 5
3. A committee of 3 persons is to be constituted from a group of 2 men and 3 women. In how many ways can this be done? How many of these committees would consist of 1 man and 2 women? 5
4. Find the least upper bound of $\{2, 9\}$ and greatest lower bound of $\{60, 72\}$ for the Poset : 5
 $(\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}, |)$.
5. Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that R is an equivalence relation. 5

SECTION—B

6. Determine the following graph is :

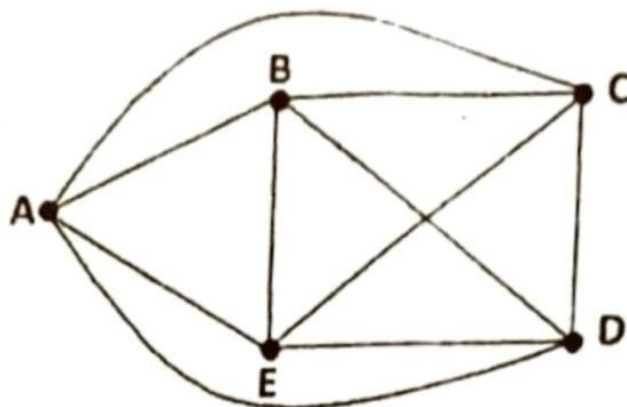
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(a) Hamiltonian, if yes, then find the Hamiltonian Cycle.

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(b) Eulerian, if yes, then find the Euler Cycle.



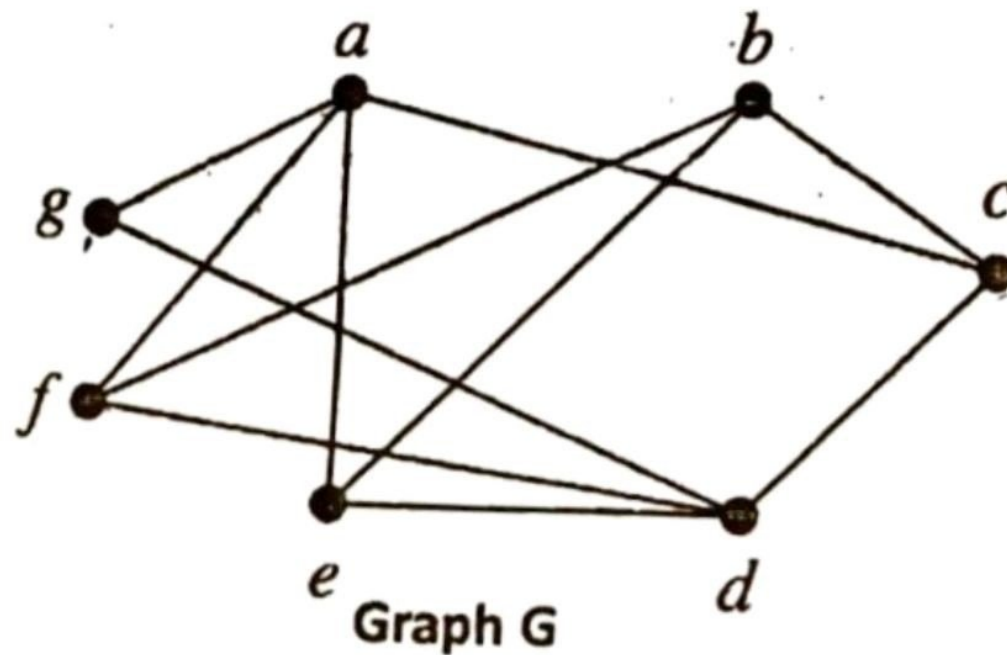
7. State and prove the Lagrange theorem.

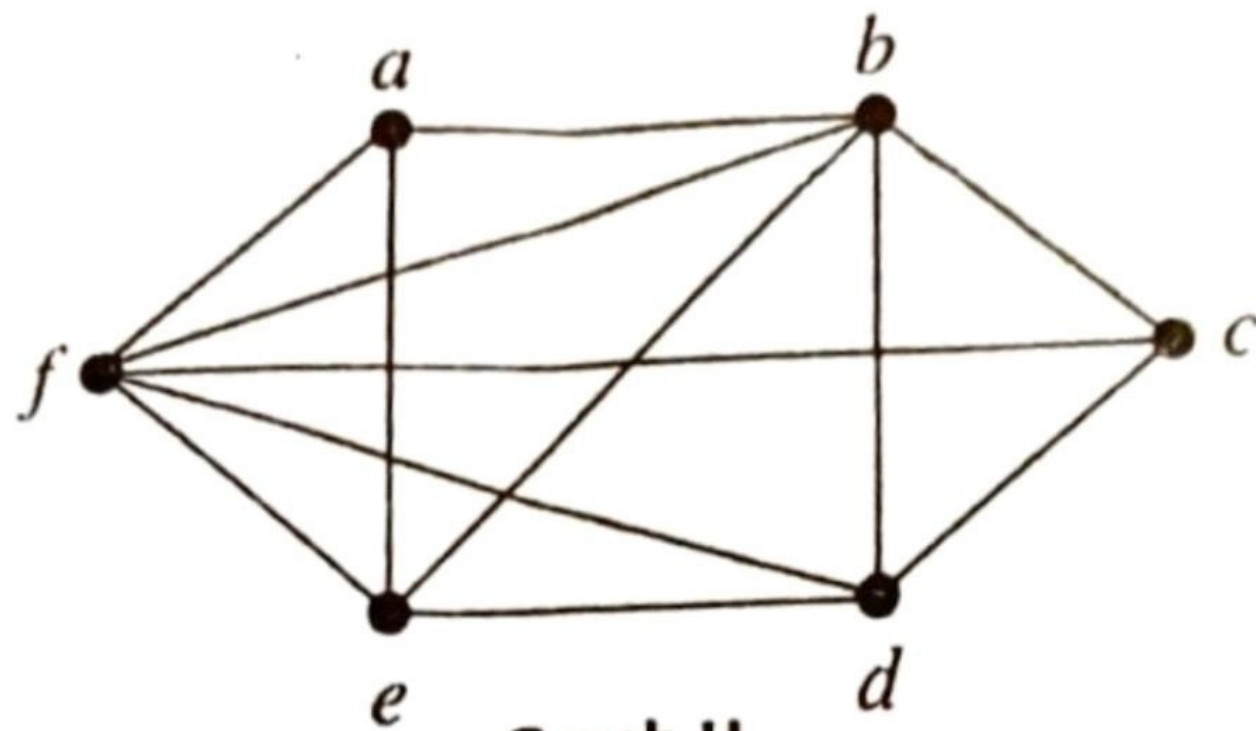
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8. Show that the set $\mathbb{Q}^+$ of all positive rational numbers forms an Abelian group under the operation * defined by $a * b = ab / 3$; $a, b \in \mathbb{Q}^+$.

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- operation defined by $a \cdot b = ab / 6, a, b \in \mathbb{Q}$.
9. Convert the expression $(p \vee q) \wedge (\neg p \vee r)$ into its equivalent Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF). 5
10. Use graph colouring theorem to determine whether the following two graphs, G and H, are bipartite?





Graph H

SECTION—C

SECTION—C

11. Answer briefly the following questions : $10 \times 2 = 20$

- (a) Draw a complete graph with five vertices.
- (b) Define the Generating function.
- (c) What is Graph isomorphism?
- (d) How many reflexive relations are there on a set with n elements?
- (e) Define the Coset.
- (f) Define a Lattice.
- (g) What is a Monoid?
- (h) What is adjacency matrix?
- (i) Define modus tollens rule.
- (j) Define the Surjection.

Roll No.

Total Pages : 4

11065/NJ**D-18/2111****DISCRETE MATHEMATICAL STRUCTURES**

Paper-205

Semester-III

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. What do you mean by spanning tree? Explain Kruskal's method to find minimum spanning tree in a graph with example. 5

2. Solve the recurrence relation : 5

$$a_n = 13a_{n-1} - 22a_{n-2} \text{ for } n \geq 2 \text{ if } a_0 = 3 \text{ and } a_1 = 15.$$

3. Draw the graph which is : 5

(a) Euler but not Hamiltonian

(b) Hamiltonian but not Euler.

4. Consider the relation : 5

$$R = \{(a, a), (a, b), (b, c), (c, c)\} \text{ on the set } A = \{a, b, c\}.$$

Find : reflexive(R), symmetric(R), transitive(R).

5. Which elements of the POSET $(\{2, 4, 5, 10, 12, 20, 25\}, |)$ are maximal and which are minimal? 5

SECTION—B

6. Construct the truth table for the compound proposition $(p \rightarrow q) \rightarrow (q \rightarrow p)$. 5

7. By using Mathematical induction prove that : 5

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

8. Explain the Lagrange theorem. 5

9. Find the values of the Boolean function represented by : 5

$$F(x, y, z) = xy + \bar{z}.$$

10. Design circuit diagram of Boolean expression $F = \bar{A}B + A\bar{B}$ using NAND gate. 5

SECTION—C

11. Answer the following questions : 10×2=20

- (i) Give an example of a nonplanar graph.
- (ii) Define Boolean algebra.
- (iii) What is recursion?

(iv) What are the equivalence classes of an equivalence relation?

(v) Give an example of a semi group without an identity element.

(vi) Define the chromatic number of a graph.

(vii) What is the number of vertices in a tree with n edges ?

(viii) Draw truth table of NOR Gate.

(xi) What is a Bipartite graph?

(x) Define the Predicate.

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DISCRETE MATHEMATICAL STRUCTURES (CPE-205)

Semester 3rd

Time: 3 Hours

Maximum Marks: 50

Note: Attempt *three* questions each from Section A and B. Section C is compulsory.

In Section A & B, each question carry *five* marks.

In Section C, each question carry *two* marks

SECTION – A

(3 * 5 Marks=15 Marks)

1. How the graphs are represented in the computer memory? Discuss any two methods.
2. Solve the recurrence relation for Fibonacci series.
3. Explain the cutpoints and bridges in the graph.
4. Find the reflexive closure and the symmetric closure of the relation R
 $R = \{(1, 2), (2, 3), (2, 4), (3, 1)\}$ on the set $\{1, 2, 3, 4\}$.
5. Determine whether these posets $(\{1, 2, 3, 4, 5\}, |)$ and $(\{1, 2, 4, 8, 16\}, |)$ are lattices?

SECTION – B

(3 * 5 Marks=15 Marks)

6. Construct the truth table for the compound proposition $(a \rightarrow b) \rightarrow (b \rightarrow a)$.
7. By Using Mathematical induction prove that

$$1 + 2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2}$$

8. Explain Lagrange theorem.
9. Use K-maps to minimize the sum-of-products expansion $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$
10. Design circuit diagram of Boolean expression $F = \bar{A}B + A\bar{B}$ using NOR gate.

SECTION – C

(10* 2 Marks=20 Marks)

11.

- a. Give an example of a planar graph.
 - b. Define Euler graph.
 - c. What is trail?
 - d. Give an example of a semi group without an identity element.
 - e. What is Hasse Diagram?
 - f. What is transitive relation?
 - g. What is one to one function?
 - h. Draw truth table of NAND Gate.
 - i. What is a generating function?
 - j. Define recursion.
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DISCRETE MATHEMATICAL STRUCTURE-205

(Semester-III)

Time : Three Hours]

[Maximum Marks : 50

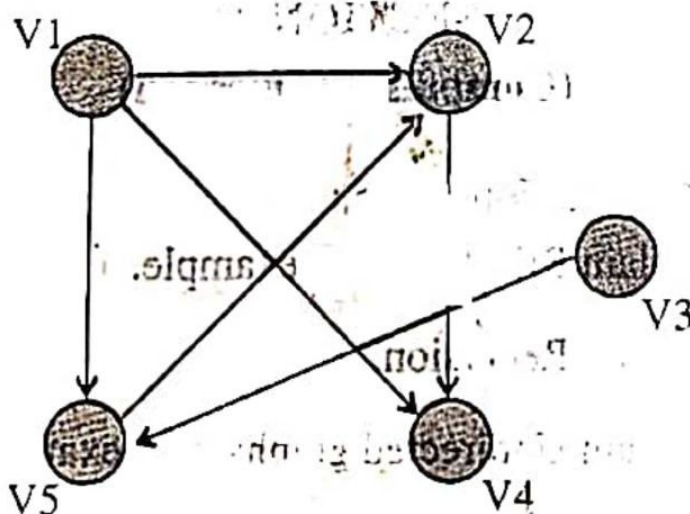
Note : Attempt *three* questions each from Section A and B carrying 5 marks each, and the entire Section-C consisting of *ten* short answer type questions carrying 2 marks each.

SECTION-A

I. Explain equivalence relation and partitions with example. (5)

II. Solve the recurrence relation $F_n = 5F_{n-1} - 6F_{n-2}$ where $F_0 = 1$ and $F_1 = 4$. (5)

III. Define Adjacency matrix. Give the adjacency matrix representation of directed graph given below : (5)



IV. Explain Hamiltonian path and Hamiltonian circuit with example. (5)

V. Explain Generating function and Graph isomorphism. (5)

SECTION-B

✓ VI. Explain Homomorphism of groups with example. (5)

✓ VII. Show that Boolean expression
 $(z' \vee x) \wedge ((x \wedge y) \vee z) \wedge (z' \vee y)$ and $(x \wedge y)$
are equivalent. (5)

✓ VIII. Explain Groups and Subgroups with examples. (5)

IX. Give the circuit diagram for $f(x, y, z) = xy' + y'z$. (5)

✓ X. Discuss Lagrange theorem and its applications in brief. (5)

SECTION-C
(Compulsory Question)

XI. Answer all the following :

- (a) Explain Bijection with example.
- (b) Explain Recursion.
- (c) Explain Undirected graphs with example.
- (d) Give Euler's formula.

✓ (e) Explain Injection with example.

✓ (f) Give an example of a finite group.

✓ (g) Give *two* applications of graphs.

✓ (h) Explain Incidence matrix.

✓ (i) Define Vertex and Degree in a graph.

✓ (j) Represent the following statement in Predicate calculus :

'Nobody likes taxes.'

(10×2=20)